

An Extensible Framework for Internet Booking Application Based on Rule Engine

Yong Zhang

Research Institute of Information
Technology, Tsinghua University
Beijing, China
zhangyong05@mail.tsinghua.edu.cn

Jie Zhao

Research Institute of Information
Technology, Tsinghua University
Beijing, China
hnsqzj@126.com

Chunxiao Xing

Research Institute of Information
Technology, Tsinghua University
Beijing, China
xingcx@tsinghua.edu.cn

Abstract—The bloom of Internet and the convenience of e-Commerce bring a lot of demands of Internet booking applications. There are two challenges in Internet Booking Applications: high volume concurrent processing and complex business processing. Facing these two challenges, we designed an extensible framework for Internet booking applications, which includes an Internet Booking Engine, a rule engine, and three kinds of servers. The architectures of e-Commerce applications usually consist of web server, application server and database server. The efficiency of the system is deeply influenced by the distribution of the functions among these servers and the communication methods between them. Three different architectures were designed and implemented. The Internet Booking Engine includes two kernel online booking modules: searching tickets and booking tickets. While dealing with complex and flexible business logics, the rule engine can reason with the business facts to get the result and take the corresponding action for specific applications by following the business rules. In vertical and horizontal directions, Internet Booking Engine can be extended to different Internet booking applications.

Keywords- Architecture; High Volume Computing; Internet Booking Engine; Rule Engine

I. INTRODUCTION

Web application has become the most popular applications on Internet [1]. The e-Commerce based on Web has already become a basic tool in daily life especially for young generation. In 2008, the transaction volume of TaoBao reached 99.96 billion Yuan, increased by 131% over last year, which was 1% of the total retail sales of China [2]. During the Spring Festival 2009, the total transaction is increased by 60% over the last year, with 0.3 billion Yuan per day [3]. Poor performance of Web-based systems can adversely impact the profitability of enterprises that rely on them. As a result, effective performance testing techniques are essential for understanding whether a Web-based system will meet its performance objectives when deployed in the real world [4]. Internet booking applications have the features of high-concurrency, real-time ability, high-reliability and security. If the general technologies are directly applied to these systems, the performances will not be guaranteed, which may result in disaster. For example, the Olympic Booking site was down in 2007.

The railway in China has the unbeatable position in increasing domestic demand to stimulate economic growth.

For the moment, the ticket system of China railway consists of separated systems running in local bureaus, which cannot satisfy the rapid development of railway. Because of the high population of China, a centralized ticket system for the whole railway is a high-volume, real-time and concurrent transaction system. According to the MOR's web site, there were 1.3 billion passengers in 2007 and the number is increased by 8.6% over 2006; there are 1.45 billion passengers in 2008 and the number is increased by 10.6% over 2007 [5]. With the development of economy, the transportation requirement of China's railway is increasing quickly. On the other hand, it is difficult to buy tickets especially during the Spring Festival and the golden tourism seasons. If the customers can buy tickets from Internet, it will save them a lot of time. How to support Internet booking has already become an urgent requirement of railway ticket system, the key issue is how to process the high-volume, real-time and concurrent transactions. We investigated it under the support of the funding of MOR of China.

The characteristics of a ticket system are quick changing, massive storage, high-volume requests and flexible management, which make the business logics very complex. In general, the business logics are distributed to separate modules in the traditional ticket systems. If the market requirements or the policies are changed, it has to maintain the system by modifying the codes, which will cost a lot of human resources and money. Therefore, it is easy to maintain and extend the system if the rules are extracted from the applications by using rule engine.

This paper is organized as follows: section 2 introduces the related technologies and tools; the system functions, architecture and performance analysis are described in section 3; section 4 presents the design of the rules of a train ticket system; section 5 gives an extensible framework of Internet booking applications using IBE as core and based on rule engine, and discusses the extending mechanisms. The last section is the summary and gives the future work.

II. RELATED TECHNOLOGIES

A. Theory of Performance Analysis

During the development of Web applications, the performance test is becoming more and more important. Performance analysis is an important part of performance test. During the performance analysis, many theoretic

analysis methods can be used, such as Petri Net, Statistics, Queuing Theory, Markov Theory, and so on.

Petri Net was invented by Carl Adam Petri in 1962 during the research on mathematic model of information systems and inter-relationships [6]. The advantages of Petri Net are theory strictness and intuition in graph modeling, which makes it to be applied in many areas, especially in describing the complex logic relationships between systems and processes such as concurrency, competition, synchronization and sequence [7].

By explicitly defining the relationships between transactions and conditions, the behaviors of the system can be described. Petri net can be used to analyze the advantages and disadvantages (such as dead-lock) in the system [8].

While modeling with Petri Net, it is critical to draw the network graph and analyze it. There are many tools to support the design and analysis. WinTTPN is a modeling tool issued in 1997 by University of Illinois. With the results from WinTTPN, the bottlenecks in the system can be found.

B. Rule Engine

Rule engine is an artificial intelligence software system that is used to manage and automatically implement business rules. The main functions include storing, classifying and managing rules, validating the consistency of rules, connecting the rules with the applications [9]. Rule engine originates from the rule-based expert systems and inference engines. It can be embedded in the application, and use the specified filtering conditions in the rules to decide if it could match the running conditions to execute the actions [10] [11].

JBoss Rules is the rule engine of JBoss application server. This engine is implemented in Java and has OO interface, so it is easy to be integrated into the softwares using Java. In JBoss Rules, a rule is consisted of a condition and a result. JBoss Rules supports the declarative programming supporting rule, that is, user defines the XML file declaration and rules, JBoss Rules interprets and executes rules [12].

C. Software Architecture

Software architecture includes system components, connectors and constrains; reflects the requirements of stakeholders; and displays the system how to meet these requirements. The design method of architecture is the model of how to meet the functional requirements and confirm to certain constraints of non-functional requirements [13].

The architectures of Web applications are mainly based on MVC, that is, Model-View-Control. The application's business can be split into three parts, which can be implemented by three kinds of servers: Web server, application server and database server. The distribution of business among these servers and the communication between them will influence the efficiency of the system deeply.

III. INTERNET BOOKING ENGINE

There are many kinds of tickets in Internet booking applications, such as train tickets, flight tickets, movie tickets, coupons and hotel rooms. By analyzing the operations on these tickets, we found that searching and booking are the two fundamental operations that influence the system performance heavily. Therefore, we designed an Internet Booking Engine (IBE) to deal with these two operations. It is easy to tackle with different tickets by extending IBE, which will be explained in section 5.

A. IBE Functions

As an engine, IBE only needs to implement searching tickets and booking tickets, and provides key services to support the complex business of different applications in the future. There are two kinds of users: administrators and customers. Administrators can manipulate on tickets and customers. Customers can search tickets, book tickets and return tickets.

Although this engine's functions are simple, it has already included the operations on tickets and users. To extend this engine, a specific ticket generating system can be developed according to specific business such as train tickets, flight tickets and hotel rooms. For example, train ticket generating system can be developed according to train no, date, origin/destination station and seat types. Because the ticket generating is not necessary to be real-time, the tickets can be generated independently and imported into the ticket database. For different businesses, different rule sets can be plugged into IBE and support complex business through rule engine. For example, the calculation of ticket price can be done in the rule engine.

B. Architecture of IBE

By distributing the business functions among Web servers, applications servers and database servers, we design three architectures as shown in figure 1.

Architecture 1: Three servers are located on one physical server. The requests are passed from the Web page to the application server through a messaging middleware. Then the application server calls the lower database access programs to operate the data. Because all of the servers are located on one physical server, the communications between them are very fast, but they compete for the resources.

Architecture 2: There are two physical servers. One hosts the Web server and the application server processing simple business logics, the other hosts the application server processing complex business logics and database server. The requests are passed between these two physical servers through a messaging middleware.

Architecture 3: There are two physical servers. One hosts the Web server, application server processing all of the business logics, the other hosts only the database server. The database can be configured and accessed by JDBC.

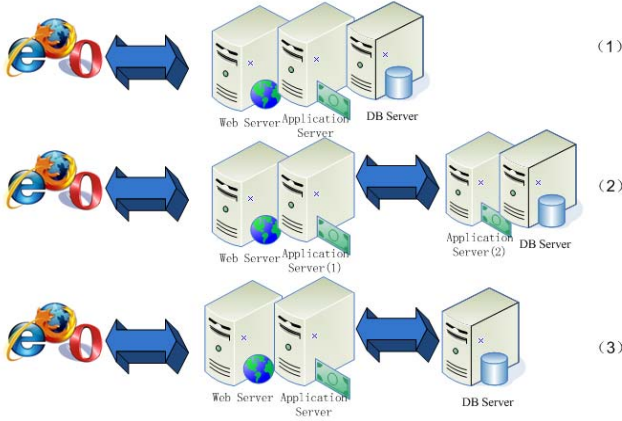


Figure 1. Three IBE architectures

C. Performance Analysis

We implemented three IBE prototypes based on the three architectures described above, and extended them to Internet booking applications for train tickets. Through WinTTPN, we found that these three architectures are bounded, secure and no dead-lock. The bottlenecks of architecture 1 and 2 are the entrance program of database services. The bottleneck of architecture 3 is the part to call the remote database server.

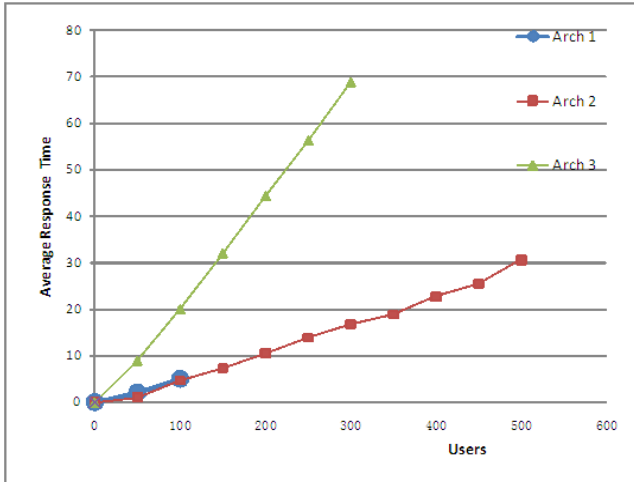


Figure 2. The performance testing on the applications of three architectures

Loadrunner is used to simulate the users' concurrent booking requests on the applications to test the performances of them relatively. Figure 3 shows the result of experiments. The average numbers of corresponding time units and concurrent users are used to evaluate the applications.

From figure 2, the response time of architecture 3 is obviously longer than the times of architecture 1 and 2; the curves of architecture 1 and 2 are almost overlapped, but the extensibility of architecture 1 is less than architecture 2.

During the whole development, by doing analysis and comparison, we have the following experiences. Based on the features of Internet booking applications, we suggest

that the kernel booking functions and database server should be put on one server, and the non-kernel and complex business logics and non real-time data operations should be put on the application server. At the same time, the complex business logics are extracted and put into rule engine for easy maintaining. The details of extensibility can be found in section V.

IV. DESIGN OF TICKET RULES

In the ticket system, the complex and changeable businesses include ticket price calculation, ticket assignment and discount strategy formulating. Ticket price calculation of train tickets is used as an example. For the convenience of description, we did some simplifications. The business rules are as following:

- price of hard seat on fast train = α * distance
- price of soft seat on fast train = β * distance
- price of hard sleeper on fast train = γ * distance
- price of hard sleeper on express train = σ * price of hard sleeper on fast train

There are parts of the rule descriptions in JBoss Rules for above rules:

```

rule "COMMON"
no-loop true
when
    p:Ticket(type == Ticket.COMMON)
then
    p.setRate( $\alpha$ );
    price = p.setRate*distance;
    modify(p);
end
rule "EXPRESS_HARD_SLEEPER"
no-loop true
when
    p:Ticket(type==Ticket.EXPRESS_HARD_SLEEPER)
then
    p.setRate( $\gamma$ * $\sigma$ );
    price = p.setRate*distance;
    modify(p);
end

```

V. THE EXTENSIBLE FRAMEWORK OF INTERNET BOOKING APPLICATIONS

Based on the experiments, an extensible framework of Internet booking applications with IBE as kernel based on rule engine is provided to solve the contradiction between fast business processing and flexible queries (figure 3). This framework includes three kinds of servers (Web server, application server, and data server), an IBE and a rule engine. These can be replaced by clusters to improve performance. Web server receives the request from Web client, and passes it to application server that does complex business logics and non real-time operations such as ticket generation. Rule Engine extracts the complex business logic from application sever to facilitate the maintenance of complex businesses. The data server receives the request through communication middleware. The functions in the request include kernel functions such as searching ticket and

booking ticket, and non-kernel functions. Because the kernel functions are high volume, it requires database supporting high volume transactions. The non-kernel functions include complex queries. It is suggested to implement the functions by using memory-database, Cache and memory-structures.

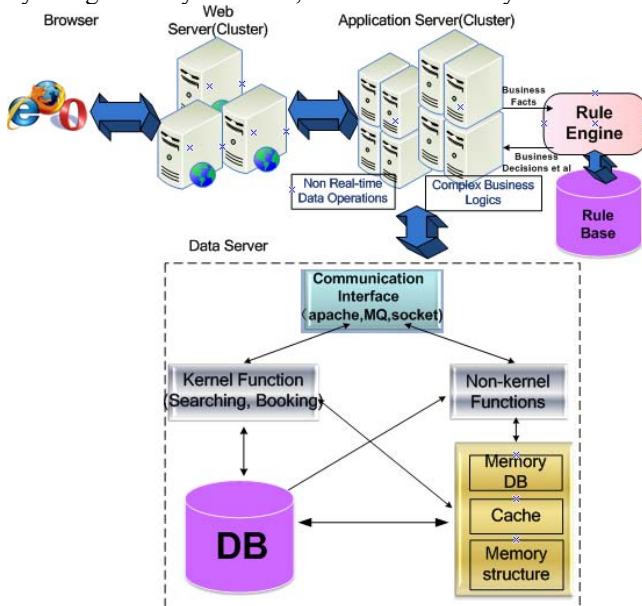


Figure 3. An extensible framework of Internet booking applications with IBE as kernel based on rule engine

The usage of rule engine changes the manner of software development, simplifies the complexity of software system, improves the readability and maintainability of code with complex logics, and separates the business decision logics of managers and technology decisions of application developers [14, 15]. Therefore, the IBE based on rule engine is an extensible system, which can be extended in vertical and horizontal directions.

For simplicity, IBE only implements the ticket price calculation. On vertical direction, more rules of train tickets can be extracted, transformed to rule sets and stored in the rule base, which will enrich the engine's functions to simplify the development and maintenance.

IBE can also be extended on horizontal direction. For example, if the rules of train ticket price are changed to airline's, this engine can be used in airline ticket booking system. However, the extensibility of this engine is beyond the booking systems. Because the rule engine decreases the coupling of logic and code, the modification and maintenance of business logic are dynamic. It is easy to change the business logics. If the ticket rules are replaced by credit card rules or market discount rules, this engine can be extended to bank loan system or market discount system.

VI. SUMMARY AND FUTURE WORK

This paper discusses the architecture design, prototype implementation and extensibility of Internet booking applications based on IBE. By comparing the relative performances of three architectures of Internet booking

applications, we summarized the advantage and disadvantage of the architectures, and gave an extensible framework of Internet booking applications with IBE as kernel based on rule engine. This framework can be extended vertically and/or horizontally to formulate different booking systems.

The future work will focus on the improvement of performance by considering operation systems and hardware, and optimization of rule engine by finding high efficient matching algorithms of ticket rules.

ACKNOWLEDGMENT

This work is supported by the MOR-Tsinghua Research Program under Grant No. J2008X009, and the National High-tech R&D Program of China under Grant No. 2009AA01Z143.

REFERENCES

- [1] Haiyan Wu, Chunyu Miao, Qixin Liu, and Fangcheng Sun, "The Research of Web Application Security Testing," Computer Security, vol. 4, 2008
- [2] www.stnn.cc (2009, Feb. 12). The transaction volume of TaoBao increased by 131% over last year. Available: http://www.singtaonet.com/chinafin/200902/t20090212_977351.html
- [3] Sohu (2009, Feb. 05) During the Spring Festival 2009, the total transaction of TaoBao is 0.46 billion per day. Available: <http://it.sohu.com/20090205/n262067073.shtml>
- [4] Mahnaz Shams, Diwakar Krishnamurthy, and Behrouz Far, "A Model-Based Approach for Testing the Performance of Web Applications," ACM New York, NY, USA, vol. 6, Nov. 2006, pp. 54-55.
- [5] Ministry of Railway of China. Available: http://www.china-mor.gov.cn/zwgk/zwgk_tjxx.html
- [6] Ke Hu, and Zhuanghui Guo, "Petri Nets and The Application on Manufacturing Systems," System Simulation Technology, vol. 4, 2006, pp. 222-223.
- [7] Xikun Pan, Tong Li, and Qing Liu, "A Hierarchical Model of Petri Net and A Modeling Tool for Its Design" May 1995.
- [8] Marco Ajmone Marsan, Gianni Conte, and Gianfranco Balbo, "A class of generalized stochastic Petri Nets for the performance evaluation of multiprocessor systems," ACM Transactions on Computer Systems, vol.2, 1984, pp. 93-122.
- [9] Xiaojun Tao, "Research and Implementation of Using Rules Engine Technology in the Enterprise Service Exploitation," M.S. thesis, Dept. Computer Center, East China Normal University, Shanghai, China, 2007.
- [10] Mingyang Jiao, and Qingping Tan, "Research on Java Rule Engine Technology," Computer Information Technology, vol. 3, 2006, pp. 41-43
- [11] Jinlong Liu, "Research and Implementation on Efficiency Optimization of Pattern-matching in Drools Rule Engine," M.S. thesis, Dept. Software Project, Southwest Jiaotong University, Chengdu, China, 2007.
- [12] Fang Guo, and Jianjun Bai, "JBoss Rules - A Rule Engine Based on Rete Algorithmm," Computer Era, China, vol. 1, 2008, pp.8-10
- [13] Hong Mei, and Junrong Shen, "Progress of Research on Software Architecture," Journal of Software, China, Vol.17, No.6, June 2006, pp.1257-1275
- [14] Gao Huang, "The Research of the Rule Engine of the Business Rule Management System-ILOG JRule," M.S. thesis, Information Engineering University, Zhengzhou, China, 2006.
- [15] Ji Liu, "The Research of Application Using Rule Engine in Business Logic Layer," M.S. thesis, Dalian Maritime University. Dalian, China, 2007.